# MAS508-G Control Theory (Reguleringsteknikk)

# Lecture 1: Introduction to Control Systems

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Leader, Robotics and Automation Group

https://www.uia.no/english/research/research-groups/engineering-and-science/robotics-and-automation.html

# Course Information

- **Credits:** 7.5,  Master's Program in Mechatronics
- **Teaching methods**
  - Lecture: 3 hours, Friday 10:15-13:00, @D3053, (Jing Zhou)
  - Tutorial/Lab: 3 hours, Friday 13:15-16:00, @ D2036, (Muhammad Faisal Aftab)
  - Project: Weeks 43-46, (Muhammad Faisal Aftab/Jing Zhou)
- **Mandatory assignment** with submission: Week 38
- **Project report:**
  - Submission: Week 46
  - Presentation: Week 47
- **Grading**
  - Project report + presentation (40 % contribution to final assessment)
  - Final exam: 4-hour individual open-book written exam (60 % contribution to final assessment)

Timetable: https://tp.educloud.no/uia/app/schedule?semester=26h&scheduleType=course&filterOpen=true&summary=true&pastWeeks=false&tab=calendar&course=MAS508%C2%A41

# Course Prerequisite

- MAS239 (UiA) / similar course in feedback control systems
- System modeling using
  - Differential equation
  - Transfer function
  - Block diagram
- System responses: $1^{st}$ and $2^{nd}$ order systems
- PID control
- Frequency response (Bode plots)
- Matlab/Simulink

# Course Contents and Teaching Plan

- Introduction of Control Systems
- System Modeling
  - Theoretical modeling
  - State-space modeling
  - Experimental modeling
  - Linearization
- State Space Design
  - Time response
  - Controllability
  - Controller design (pole placement and LQR)
  - Observability
  - Observer design (pole placement and LQR)
  - Output feedback control
  - Linear quadratic regulator (LQR) & Linear quadratic Gaussian (LQG)
- Stability
- Design via Frequency Response (Lag, Lead, Lag-Lead)
- Advanced control

# MAS508

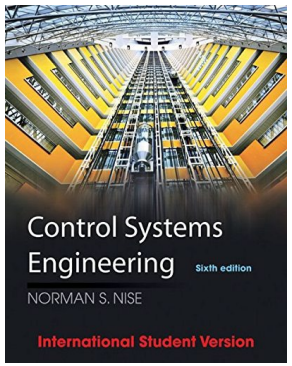
## Labs

- Matlab/Simulink
- Speedgoat real time simulation



## Textbook:

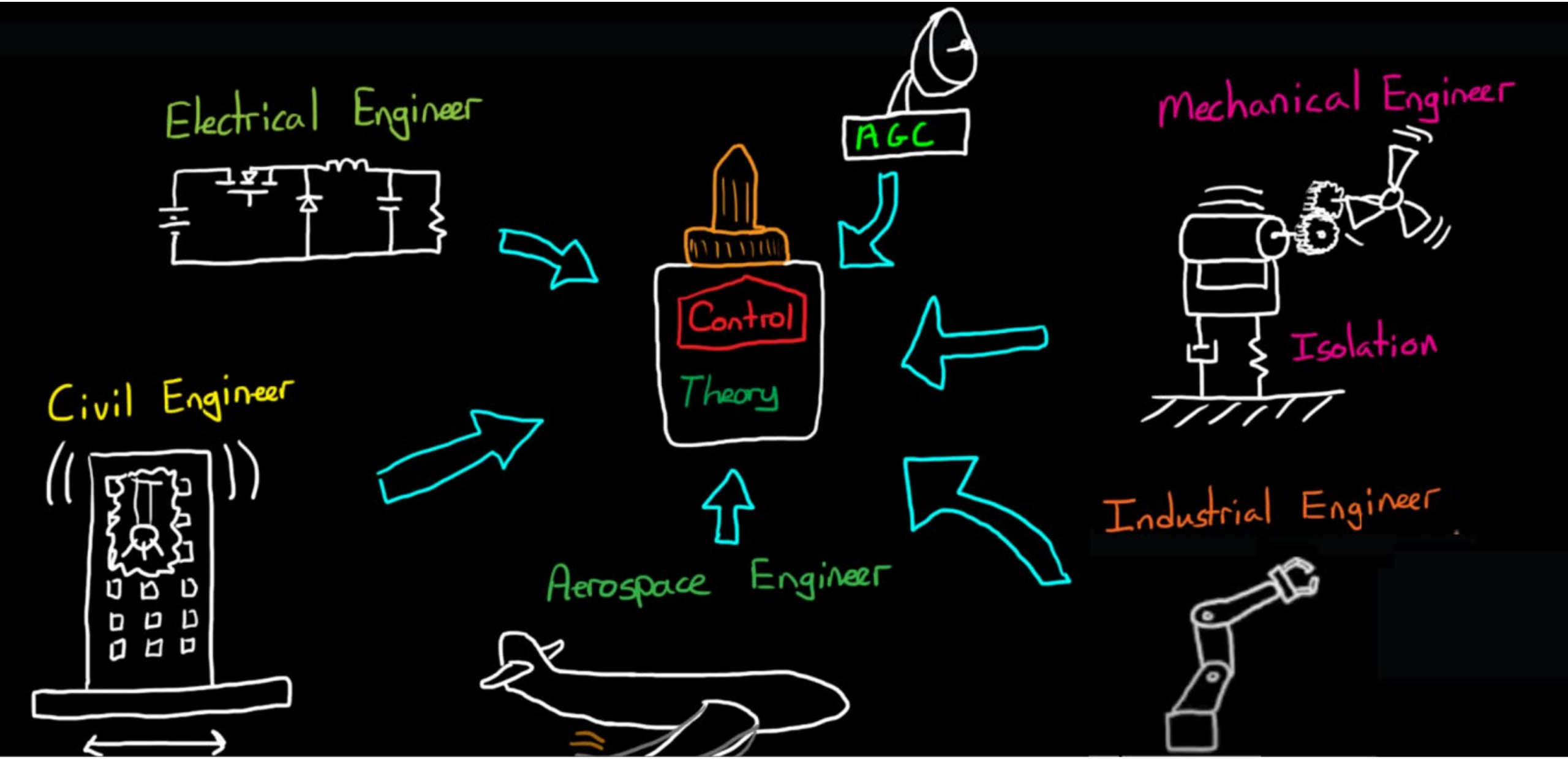
- Control System Engineering, Norman S. Nise (all editions)
- The copy of chapters 3 and 12 is available on Canvas.



# Control is Everywhere

- The modern world is driven by control systems
- Modern engineers must understand control systems for all disciplines




Electrical Engineer
AGC
Mechanical Engineer
Control
Theory
Isolation
Civil Engineer
Industrial Engineer
Aerospace Engineer



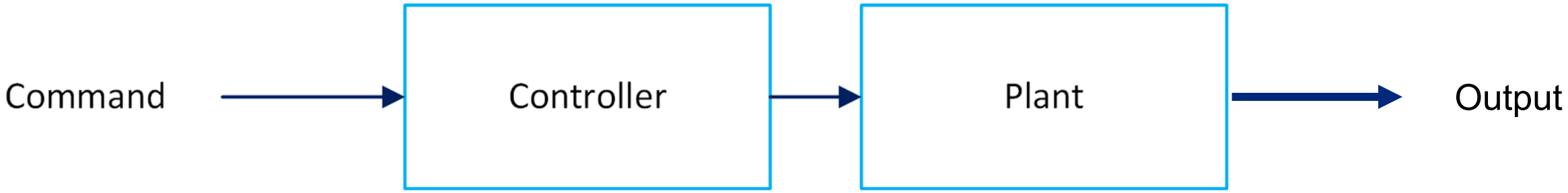

Control systems are responsible for determining the appropriate system response to measurements from the environment.

# Open-loop Control

- Controllers generally attempt to drive a *plant* to a desired *state*
- Open-loop controller
  - Changes the output from the controller based on a **model** of the plant
  - Only used to control simple systems with known dynamics
  - Does not account for disturbances



# Open-loop Control




**Video**

https://se.mathworks.com/videos/understanding-control-systems-part-1-open-loop-control-systems-123419.html

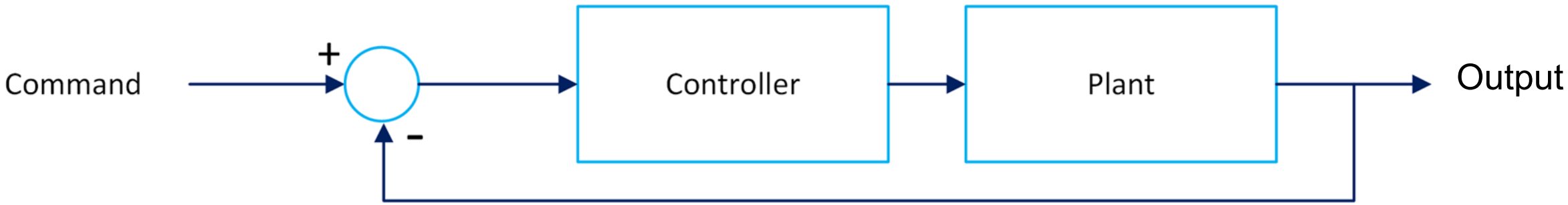
# Closed-loop Control

## Closed-loop controller

- Changes the output from the controller based on the **measured state** of the plant
- Able to compensate for both changes in command and disturbances to the system
- Most common modern approach to control



# Closed-loop Control (Feedback control)




**Video**

https://se.mathworks.com/videos/understanding-control-systems-part-2-feedback-control-systems-123501.html

# Components of feedback control

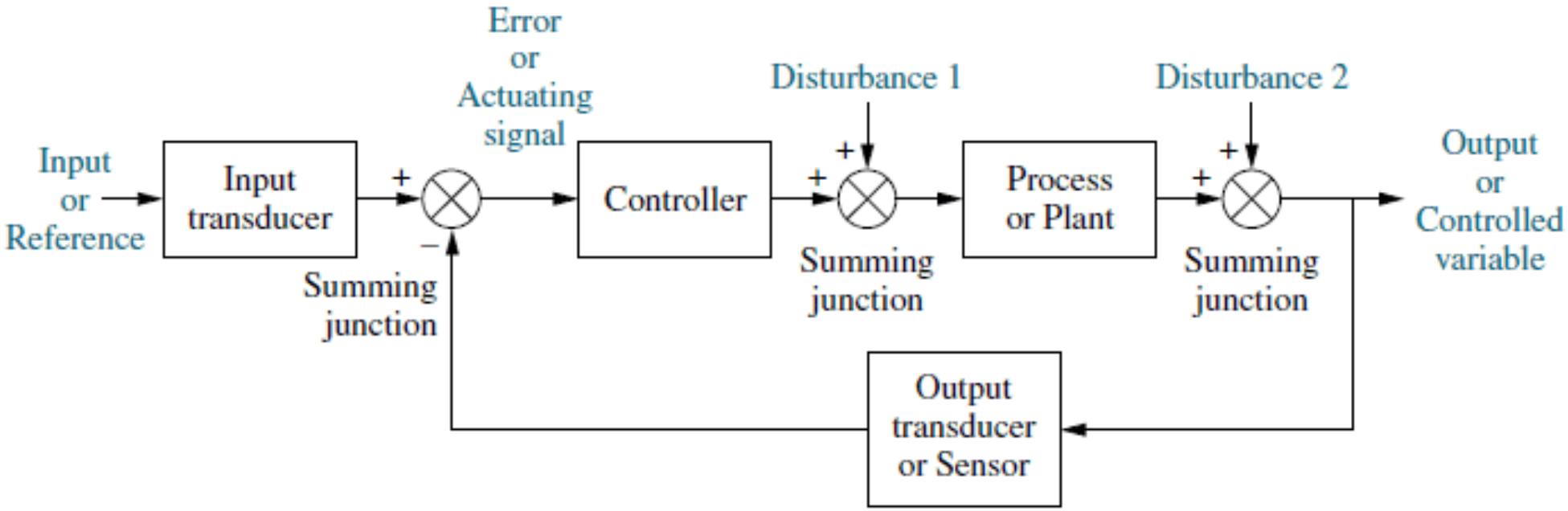



Fig. [Nise, Control System Engineering]

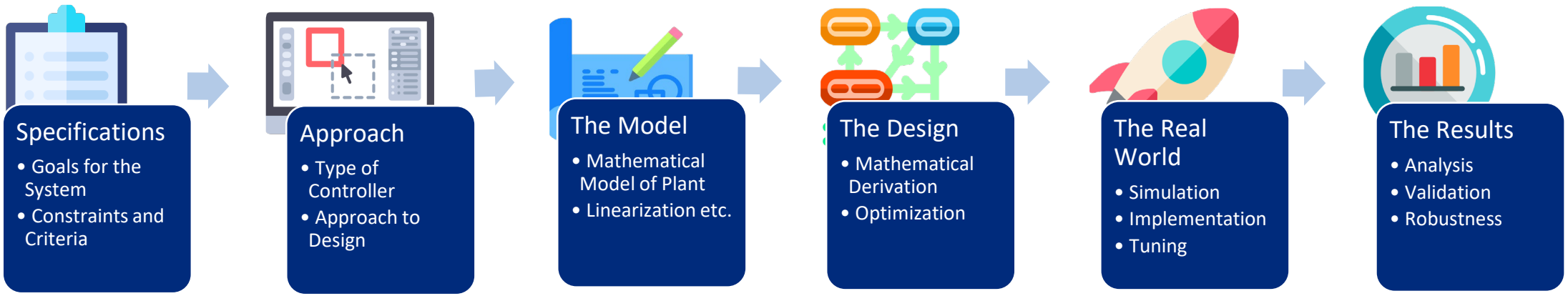
## Videos

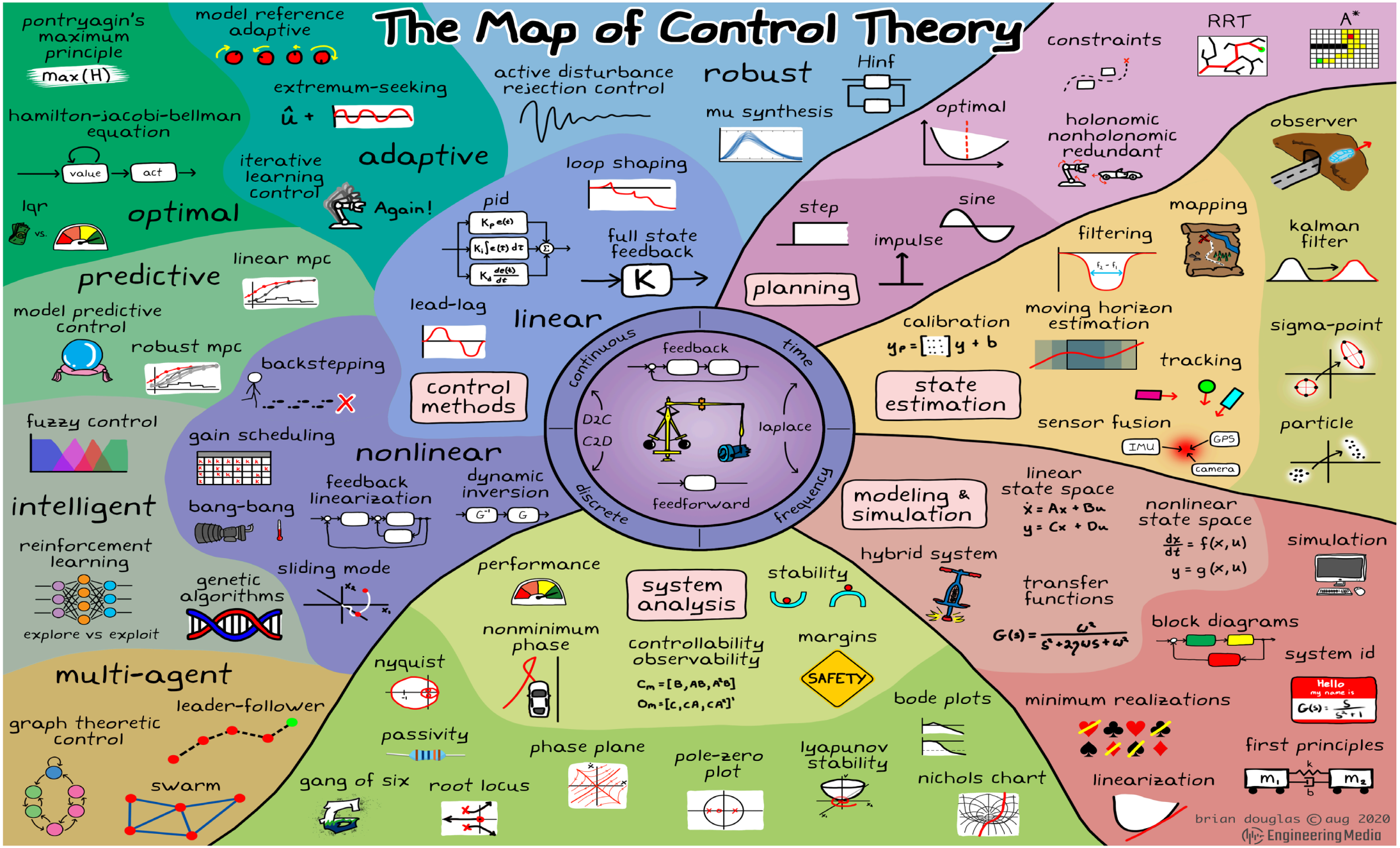
https://se.mathworks.com/videos/understanding-control-systems-part-3-components-of-a-feedback-control-system-123645.html
https://www.youtube.com/watch?v=lBC1nEq0_nk&t=470s

# Control Design

- The design process begins with WHAT you want the plant to do, and what controller makes sense based on the properties of the system.
- You then design the controller mathematically based on a system model if possible and test the proposed controller using a simulation.
- Finally, you implement the controller on the actual system and tune in order to achieve the desired response.

# Control Design




The Map of Control Theory
pontryagin's maximum principle
Max(H)
model reference adaptive
extremum-seeking
active disturbance rejection control
robust
Hinf
mu synthesis
constraints
RRT
A*
hamilton-jacobi-bellman equation
value
act
iterative learning control
Again!
adaptive
optimal
lqr
vs.
optimal
holonomic
nonholonomic
redundant
observer
loop shaping
pid
full state feedback
K
step
impulse
sine
planning
mapping
filtering
kalman filter
predictive
linear mpc
model predictive control
robust mpc
lead-lag
linear
control methods
backstepping
calibration
moving horizon estimation
state estimation
tracking
sigma-point
fuzzy control
gain scheduling
nonlinear
sensor fusion
IMU
GPS
camera
particle
continuous
discrete
time
frequency
feedback
feedforward
laplace
D2C
C2D
intelligent
feedback linearization
dynamic inversion
bang-bang
modeling & simulation
linear state space
nonlinear state space
simulation
reinforcement learning
explore vs exploit
genetic algorithms
sliding mode
hybrid system
transfer functions
block diagrams
system id
performance
system analysis
stability
nonminimum phase
controllability
observability
margins
SAFETY
nyquist
bode plots
minimum realizations
multi-agent
leader-follower
graph theoretic control
swarm
passivity
phase plane
pole-zero plot
lyapunov stability
nichols chart
first principles
gang of six
root locus
linearization
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